ETV6-Related Thrombocytopenia (Thrombocytopenia 5): Comprehensive Disease Characterization Report

Disease: ETV6-Related Thrombocytopenia (Thrombocytopenia 5, THC5) **Causal gene: ETV6** (ETS variant transcription factor 6; formerly *TEL*), HGNC:3495, chromosome 12p13.2, NCBI Gene 2120 **Key identifiers:** OMIM:616216 (Thrombocytopenia 5, THC5) · MONDO:0014536 · Orphanet (ETV6-related thrombocytopenia with predisposition to leukemia) · Inheritance: Autosomal dominant · Category: Mendelian

> This report concerns the Mendelian disorder caused by germline variants in **ETV6**. It must not be confused with similarly numbered inherited thrombocytopenias caused by other genes (e.g., *ANKRD26*-related THC2, *RUNX1*-related familial platelet disorder/THC-FPD); the defining gene here is **ETV6**.

Summary

ETV6-Related Thrombocytopenia is an autosomal dominant Mendelian bleeding/cancer-predisposition syndrome caused by germline heterozygous variants in *ETV6*, the gene encoding the master hematopoietic transcriptional repressor ETV6 (TEL). The disease was defined in 2015 when whole-exome sequencing of a family with dominantly inherited thrombocytopenia, elevated erythrocyte mean corpuscular volume (MCV), and two cases of B-cell precursor acute lymphoblastic leukemia (ALL) identified a heterozygous *ETV6* c.641C>T (p.Pro214Leu) variant segregating with disease (PMID: 25807284). The clinical phenotype is a lifelong, mild-to-moderate thrombocytopenia with red-cell macrocytosis and generally mild bleeding, but its defining and most consequential feature is a high, age-dependent, incompletely penetrant predisposition to hematological malignancy — principally childhood B-ALL, plus adult myelodysplastic syndrome (MDS) and acute myeloid leukemia (AML).

Mechanistically, pathogenic *ETV6* variants act as **dominant-negative loss-of-function alleles**. Variants cluster in the ETS (DNA-binding) and central/linker domains. The mutant protein is mislocalized from the nucleus to the cytoplasm and physically co-sequesters wild-type ETV6, abolishing nuclear transcriptional repression of ETV6 target genes. This produces two main phenotypic branches: (1) dysregulated megakaryopoiesis — an expanded but immature megakaryocyte pool with abnormal cytoskeletal organization and impaired proplatelet formation, yielding thrombocytopenia with largely preserved platelet size; and (2) a predisposed hematopoietic stem/progenitor compartment that, upon acquiring a somatic "second hit" (frequently deletion of the remaining wild-type *ETV6* allele plus cooperating mutations), transforms to leukemia. A CRISPR knock-in mouse of the recurrent human mutation adds a third, inflammation-driven branch: elevated IL-18 and IL-13 impair B-lymphoid development of multipotent progenitors, linking ETV6 dysfunction to both defective thrombopoiesis and leukemogenesis.

There is **no disease-specific drug**. Management is surveillance-based: lifelong hematologic monitoring for malignant transformation, genetic counseling and cascade testing, avoidance of immune-thrombocytopenia (ITP)-type therapies (splenectomy, immunosuppression) that are inappropriate for this inherited disorder, and — critically — carrier-aware selection of related donors for allogeneic hematopoietic stem cell transplantation (HSCT) to prevent donor-derived leukemia. The disorder is ultra-rare (only ~18 families reported by 2017), and no population prevalence figures are firmly established. Knowledge derives almost entirely from aggregated disease-level resources — pedigree case reports/series (human clinical), in vitro functional assays, patient iPSC models, and genetically engineered mouse models — rather than individual-level EHR data.

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1. Disease Information

Overview. ETV6-Related Thrombocytopenia (Thrombocytopenia 5; THC5) is an autosomal dominant inherited platelet disorder with predisposition to hematological malignancy. Patients present, usually in childhood, with decreased platelet number and a tendency to bleed, and carry germline mutations in *ETV6*, a master hematopoietic transcription factor (PMID: 28555414): *"Patients with thrombocytopenia 5 have an autosomal dominant disorder of decreased platelet number with tendency to bleed, usually presenting in childhood, and have been found to have germline mutations in ETV6, which encodes a master hematopoietic transcription factor."*

Key identifiers.

Resource	Identifier
OMIM	616216 (Thrombocytopenia 5, THC5)
MONDO	MONDO:0014536
Gene (HGNC)	ETV6, HGNC:3495
NCBI Gene	2120
Chromosome	12p13.2
Orphanet	ETV6-related thrombocytopenia with predisposition to acute lymphoblastic leukemia

Synonyms / alternative names: ETV6 thrombocytopenia; ETV6-related thrombocytopenia with or without myeloid malignancy; thrombocytopenia caused by mutation in ETV6; ETV6-related thrombocytopenia and leukemia predisposition; THC5. The gene was historically called *TEL* (translocation-ETS-leukemia).

Information source type. Disease-level aggregated resources (OMIM, Orphanet) and individual-patient/family-level primary literature (pedigree studies, functional characterizations). Not EHR-population-derived; a rare Mendelian disorder characterized through family-based gene discovery and functional studies.

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2. Etiology

Primary cause — genetic. The disease is monogenic: a germline heterozygous pathogenic variant in *ETV6* is the necessary cause. The original gene-discovery study identified *ETV6* c.641C>T (p.Pro214Leu) by whole-exome sequencing and found two additional *ETV6*-mutant families among 23 screened (PMID: 25807284): *"Whole-exome sequencing identified a heterozygous single-nucleotide change in ETV6 (ets variant 6), c.641C>T, encoding a p.Pro214Leu substitution in the central domain, segregating with thrombocytopenia and elevated MCV."*

Genetic risk factors. The causal variants themselves (Section 4) are the risk factors for thrombocytopenia. For the *malignancy* phenotype, the germline variant establishes a predisposed clone that requires a somatic second hit — the strongest driver of progression is acquisition of additional somatic lesions, most characteristically loss of the wild-type *ETV6* allele (PMID: 26102509). Modifier genes are not firmly established.

Environmental, lifestyle, and infectious factors. No specific environmental toxin, radiation, occupational exposure, lifestyle factor (smoking, diet, alcohol), or infectious agent has been identified as causing or triggering ETV6-related thrombocytopenia. This is a germline Mendelian disorder; environmental contributions are not applicable to onset. As a non-specific consideration, prior cytotoxic chemotherapy may contribute to therapy-related MDS in predisposed carriers — one reported sibling developed therapy-related MDS after childhood ALL (PMID: 40858768).

Protective factors. No genetic or environmental protective factors are established.

Gene–environment interactions. No specific gene–environment interaction is characterized. The dominant interaction is gene–somatic-mutation (two-hit): germline ETV6 loss predisposes; a stochastic/acquired somatic event completes transformation.

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3. Phenotypes

The core phenotype is **mild-to-moderate thrombocytopenia with a tendency to bleed, red-cell macrocytosis, impaired platelet function, and predisposition to hematological malignancy** (PMID: 28555414; PMID: 25807284).

Phenotype	Type	HPO suggestion	Onset	Severity / frequency	Progression
Thrombocytopenia	Laboratory abnormality / clinical sign	HP:0001873	Childhood (often congenital/ early)	Mild–moderate; high penetrance	Stable, lifelong
Red-cell macrocytosis (high MCV)	Laboratory abnormality	HP:0001972	Childhood	Co-segregates with thrombocytopenia	Stable
Abnormal platelet aggregation / dysfunction	Laboratory abnormality	HP:0003540	Lifelong	Impaired aggregation in ~90% of IT patients tested	Stable
Bleeding tendency (bruising, epistaxis, mucosal)	Symptom / clinical sign	HP:0001892	Childhood	Generally mild; often not main issue	Episodic with hemostatic challenge
B-cell acute lymphoblastic leukemia	Neoplasm	HP:0006721	Childhood	Subset; part of ~10–45% HM risk	Progressive if untreated
Myelodysplastic syndrome	Neoplasm	HP:0002863	Adult	Subset	Progressive
Acute myeloid leukemia	Neoplasm	HP:0004808	Adult	Subset	Progressive
Normal/near-normal platelet size	Laboratory feature	—	—	Distinguishes from macrothrombocytopenias	Stable

Key quantitative/clinical details. In a cohort of inherited thrombocytopenia with malignancy risk, *"Platelet aggregation was impaired in 90% of all patients, while reduced activation and granule secretion were heterogeneous"* (PMID: 40670159); genetic diagnosis was delayed ~20 years from first thrombocytopenia detection. The original family displayed the characteristic triad: *"autosomal dominant thrombocytopenia, high erythrocyte mean corpuscular volume (MCV) and two occurrences of B cell-precursor acute lymphoblastic leukemia (ALL)"* (PMID: 25807284).

Quality-of-life impact. Bleeding is frequently NOT the dominant clinical problem; it is usually mild. The principal burden on quality of life and prognosis is the lifelong cancer risk and surveillance need (PMID: 28594466: *"mutations in RUNX1, ANKRD26 and ETV6 cause congenital thrombocytopaenia, but we now know that they also predispose to haematological malignancies"*). Formal EQ-5D/SF-36 data specific to this ultra-rare disorder are not available.

4. Genetic / Molecular Information

Causal gene. *ETV6* (ETS variant transcription factor 6; TEL), HGNC:3495, chromosome 12p13.2, NCBI Gene 2120. ETV6 is an ETS-family transcriptional repressor and master hematopoietic regulator (PMID: 28555414).

Pathogenic variants. Reported germline pathogenic/likely-pathogenic variants cluster in the **ETS (DNA-binding) domain** and the **central/linker domain**, predominantly **missense** with occasional **frameshift** (PMID: 25807284; PMID: 26102509; PMID: 27663637; PMID: 40858768):

Variant (protein)	cDNA	Domain	Type	Note
p.Pro214Leu	c.641C>T	Central/ linker	Missense	Recurrent "founder-like" hotspot; LOF control in assays; predicted extra nuclear export signal
p.Arg418Gly	c.1252A>G	ETS DNA-binding	Missense	Associated with alternative splicing / exon skipping
p.Leu349Pro	—	Central	Missense	Thrombocytopenia + ALL kindred
p.Asn385fs	—	—	Frameshift	WT-ETV6 allele deleted in leukemic cells
p.Ala377Thr	—	ETS	Missense	Reduced corepressor binding
p.Tyr401Asn	—	ETS	Missense	Reduced corepressor binding
p.Leu376Gln	—	—	Missense	Hematological + solid-tumor pedigree; absent from gnomAD
p.Arg202Gly	c.604C>G	Central	Missense	**WT-like in functional assays — NOT loss-of-function** (cautionary VUS)

Variant classification (ACMG/AMP). Pathogenicity is variant-specific and requires functional evaluation. A key cautionary example: p.Arg202Gly, found in monozygotic twins with ETV6::RUNX1-positive B-ALL, behaved WT-like in localization and reporter assays and does **not** support a loss-of-function effect, whereas p.Pro214Leu failed to repress target promoters (PMID: 42479114). Rare *ETV6* variants should not be over-called as pathogenic without functional data.

Allele frequency. Pathogenic variants are essentially absent from large population databases (e.g., p.Leu376Gln not in gnomAD), consistent with pathogenic rarity (PMID: 40858768).

Somatic vs germline. The disease-defining variant is **germline heterozygous**. A somatic second hit — frequently deletion/loss of the remaining wild-type *ETV6* allele — occurs in leukemic cells: *"Leukemic cells from the proband in the second kindred showed deletion of wild type ETV6 with retention of the ETV6 p.N385fs"* (PMID: 26102509). Germline status must be confirmed in non-hematopoietic tissue (skin fibroblasts) to avoid mistaking a germline allele for somatic clonal hematopoiesis (PMID: 40858768).

Functional consequence. Dominant-negative loss of function. Mutant ETV6 is mislocalized to the cytoplasm and co-sequesters wild-type ETV6, reducing transcriptional repression (PMID: 25807284; PMID: 26102509). ETS-domain variants additionally reduce corepressor binding (PMID: 27663637).

Modifier genes, epigenetics, chromosomal abnormalities. No specific modifier genes or disease-specific epigenetic signatures are established for the germline disorder. Large-scale chromosomal abnormalities are not the primary germline lesion; however, somatic loss of 12p / the wild-type *ETV6* allele participates in transformation.

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5. Environmental Information

No environmental, lifestyle, or infectious factors are established as causes or triggers — it is a germline Mendelian disorder. As a non-specific consideration, prior cytotoxic therapy for a first malignancy may contribute to therapy-related MDS in predisposed carriers (PMID: 40858768), but no primary environmental etiology applies. **Not applicable / not available.**

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **germline heterozygous *ETV6* variant** in the ETS or central/linker domain **leads to** production of a structurally altered ETV6 protein (normal transcript/protein levels in several variants) (PMID: 25807284; PMID: 26102509).
2. The mutant protein **results in** impaired nuclear localization — ETV6 is mislocalized to the **cytoplasm**, and (dominant-negative) co-sequesters endogenous wild-type ETV6 (*inferred* to involve an exposed/created nuclear export signal, predicted in silico for p.Pro214Leu) (PMID: 25807284; PMID: 42479114).
3. Cytoplasmic mislocalization **leads to** loss of nuclear ETV6-mediated transcriptional repression; ETS-domain variants additionally **result in** reduced corepressor binding (PMID: 26102509; PMID: 27663637).
4. Loss of repression **branches** into three downstream effects:
5. **4a (thrombocytopenia branch):** Dysregulated megakaryopoiesis — expansion of megakaryocyte colony-forming units but production of **immature megakaryocytes** with abnormal cytoskeletal organization **leads to** defective **proplatelet formation** → reduced platelet release → thrombocytopenia (largely preserved platelet size) → mild bleeding (PMID: 27663637; PMID: 34019812).
6. **4b (erythroid branch):** Altered erythropoiesis **results in** red-cell macrocytosis (elevated MCV) (PMID: 25807284).
7. **4c (inflammation/lymphoid branch):** ETV6 dysfunction **promotes** sterile inflammation; elevated **IL-18 and IL-13 abrogate** B-cell development of multipotent progenitor (MPP4) cells (*demonstrated in mouse knock-in; mechanistic inference for human leukemogenesis*) (PMID: 35803545).
8. Over years, a predisposed clone **acquires a somatic second hit** — frequently **deletion/loss of the remaining wild-type *ETV6* allele** plus cooperating mutations (PMID: 26102509).
9. The second hit **leads to** clonal expansion → hematological malignancy (childhood B-ALL; adult MDS/AML) (PMID: 28555414; PMID: 26102509).

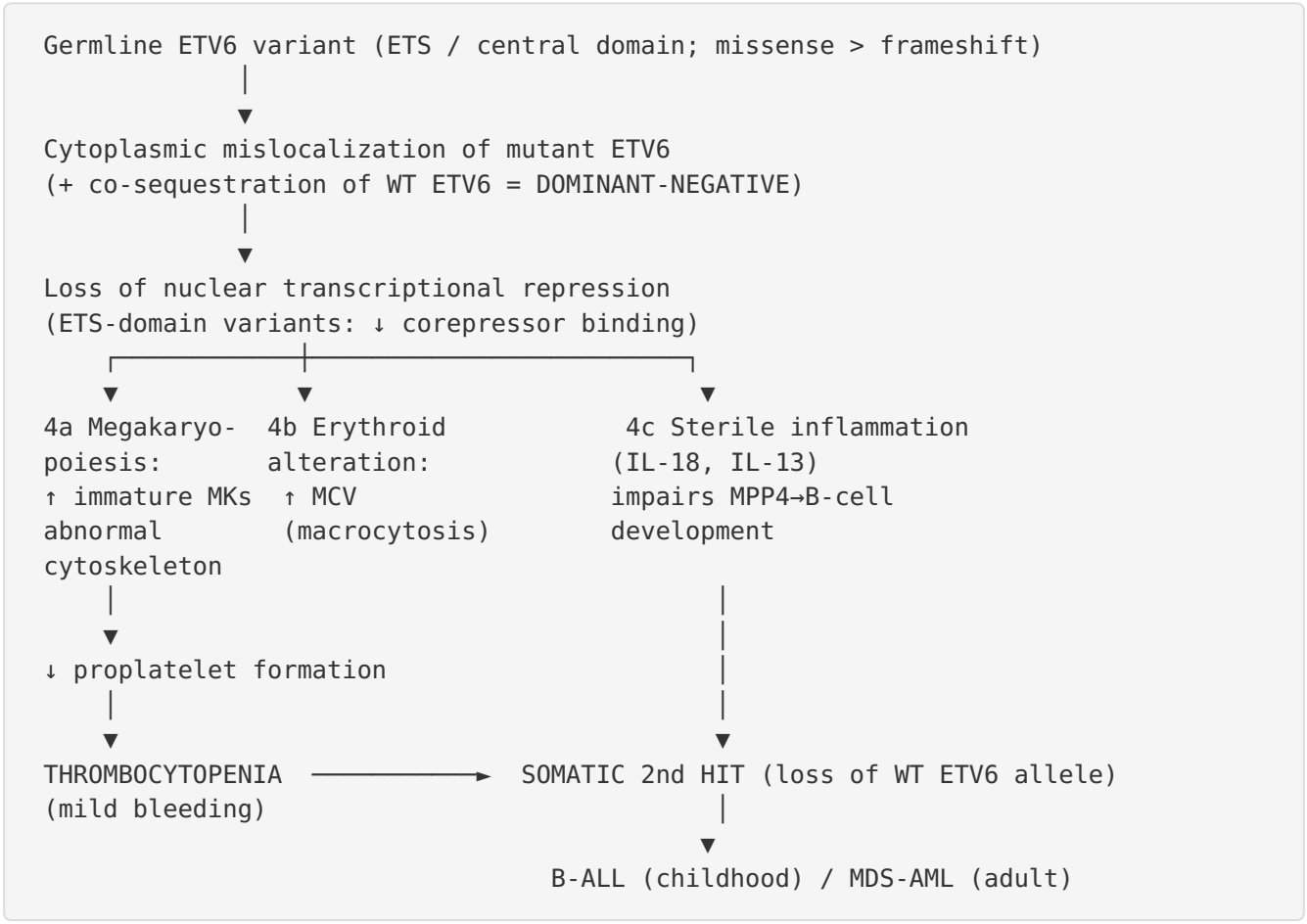
Mechanistic detail

- **Molecular pathways / cellular processes:** ETV6 performs sequence-specific **DNA-binding transcriptional repression** (GO:0000122) and regulates **hematopoietic/megakaryocyte differentiation**. Disease is driven by loss of this repressive function. Suggested GO terms:

GO:0030219 (megakaryocyte differentiation), GO:0030220 (platelet formation), GO:0006357 (regulation of transcription by RNA Pol II), GO:0050729 (positive regulation of inflammatory response — IL-18/IL-13 branch).

- **Protein dysfunction:** Dominant-negative loss of function via subcellular mislocalization (nucleus → cytoplasm) and co-sequestration of wild-type protein; ETS-domain variants impair corepressor recruitment. Suggested GO cellular components: GO:0005634 (nucleus — normal site), GO:0005737 (cytoplasm — aberrant localization).
- **Immune/inflammatory involvement:** The CRISPR knock-in mouse revealed sterile inflammation with IL-18 and IL-13 impairing lymphoid development — *"a model in which ETV6 dysfunction promotes inflammation, which adversely affects thrombopoiesis and promotes leukemogenesis"* (PMID: 35803545).
- **Cell types (CL terms):** megakaryocyte (CL:0000556), platelet (CL:0000233), erythroid lineage (CL:0000764), hematopoietic multipotent progenitor (CL:0000837), common lymphoid progenitor (CL:0000051).
- **Upstream vs downstream:** Upstream = germline variant → cytoplasmic mislocalization → loss of repression. Downstream = megakaryocyte maturation defect/thrombocytopenia, macrocytosis, inflammation-impaired lymphopoiesis, and (after second hit) leukemia.

Schematic



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7. Anatomical Structures Affected

- **Organ/system level:** The **hematopoietic system** and **bone marrow** (UBERON:0002371, bone marrow; UBERON:0002390, hematopoietic system) are the primary sites. ETV6 is required specifically for bone-marrow hematopoiesis (PMID: 9694803). Secondary involvement: peripheral blood and, in leukemia, extramedullary sites.
- **Tissue/cell level:** Affected cells are **megakaryocytes** (CL:0000556; overproduced but immature with defective proplatelet formation — PMID: 34019812, PMID: 27663637), the **erythroid lineage** (CL:0000764; macrocytosis — PMID: 25807284), and **multipotent/lymphoid progenitors** (MPP4; CL:0000837 — PMID: 35803545). Carriers show increased circulating **CD34+ progenitors** (PMID: 27663637).

- **Subcellular level:** The core defect is subcellular localization — mutant ETV6 is displaced from the **nucleus** (GO:0005634) to the **cytoplasm** (GO:0005737) (PMID: 25807284; PMID: 26102509).
- **Localization/lateralization:** Systemic hematologic disorder; lateralization not applicable.

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8. Temporal Development

- **Onset:** Thrombocytopenia typically presents in **childhood** (often congenital/early-life) and is chronic/lifelong (PMID: 28555414); onset pattern **chronic/insidious**.
- **Progression:** The thrombocytopenia itself is **stable** over the lifespan. Malignancy risk is **age-dependent**: B-ALL characteristically presents in childhood, MDS/AML in adulthood — within a single pedigree one sibling had childhood ALL and adult therapy-related MDS while another had only chronic isolated thrombocytopenia (PMID: 40858768).
- **Disease course:** Lifelong (chronic) for thrombocytopenia; the leukemic branch, when it occurs, is a discrete transformative event requiring intensive therapy.
- **Critical periods / intervention windows:** Continuous lifelong hematologic surveillance is the practical intervention window; no defined molecular "critical period" for prevention is established.

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9. Inheritance and Population

- **Inheritance: Autosomal dominant** (PMID: 25807284; PMID: 28555414).
- **Penetrance: High (near-complete) for thrombocytopenia/macrocytosis**, which co-segregate with the variant; **incomplete, age-dependent for hematological malignancy** — ~10–45% across the ETV6/RUNX1/ANKRD26 group (PMID: 40670159: *"Inherited thrombocytopenia (IT) with germline variants in RUNX1, ETV6 or ANKRD26 carries a high risk (10%-45%) of developing haematological malignancy."*).
- **Expressivity: Variable**, even within a single family (PMID: 40858768).
- **Epidemiology: Ultra-rare** — only **18 families** had been reported by 2017 (PMID: 28555414: *"Since the first description of germline ETV6 mutations, 18 families have been reported."*). No established population prevalence/incidence; no formal Orphanet prevalence class.

- **Founder effects / anticipation / mosaicism / consanguinity / carrier frequency:** No firmly established founder population, though **p.Pro214Leu recurs** across unrelated families (recurrent hotspot/founder-like allele). Genetic anticipation and germline mosaicism not established. Carrier frequency unknown (variants absent from gnomAD).
- **Demographics:** No firmly established ethnic predilection, sex bias, or geographic clustering. Age distribution: thrombocytopenia from childhood; B-ALL in children, MDS/AML in adults.

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10. Diagnostics

- **Clinical/laboratory tests:** CBC shows **isolated mild-to-moderate thrombocytopenia** with **elevated MCV (macrocytosis)** and **near-normal platelet size**; platelet aggregation studies are typically **impaired** (PMID: 25807284; PMID: 40670159). LOINC: platelet count (26515-7), MCV (30428-7).
- **Genetic testing (definitive):** Diagnosis is molecular. **Targeted panel-based NGS** covering inherited platelet-disorder genes (e.g., 59-gene panels including ETV6, RUNX1, ANKRD26) is standard (PMID: 31249973: *"we developed, verified, and evaluated a targeted, panel-based next-generation sequencing approach comprising 59 genes associated with IPD"*). WES/WGS aid gene discovery and atypical cases. **Germline confirmation on non-hematopoietic tissue** (cultured skin fibroblasts) is required to distinguish germline from somatic/clonal variants (PMID: 40858768: *"Germline testing on skin fibroblasts confirmed the presence of the same ETV6 variant in the proband and two of his siblings"*).
- **Variant interpretation caveat:** Rare *ETV6* variants require **functional assessment** (subcellular localization + transcriptional reporter assays) before being called pathogenic; p.Arg202Gly is WT-like despite occurring in leukemia twins (PMID: 42479114).
- **Diagnostic delay:** Frequently ~20 years from first thrombocytopenia detection to genetic diagnosis (PMID: 40670159).
- **Differential diagnosis:** Other inherited thrombocytopenias with malignancy predisposition (RUNX1-RT/FPDMM, ANKRD26-RT), immune thrombocytopenia (ITP), Bernard-Soulier syndrome, other inherited platelet disorders. Distinguishing features: ETV6 shows macrocytosis, near-normal platelet size, and a distinctive molecular signature (overproduced immature megakaryocytes vs. RUNX1's reduced, more agonist-responsive megakaryocytes — PMID: 34019812).
- **Screening:** Cascade testing of at-risk relatives; no population newborn screening exists.

11. Outcome / Prognosis

- **Thrombocytopenia prognosis:** Generally benign; bleeding is usually mild and *"not the main clinical issue"* (PMID: 28594466).
- **Malignancy dominates prognosis:** The ~10–45% lifetime hematological-malignancy risk (childhood B-ALL; adult MDS/AML) is the principal determinant of outcome (PMID: 40670159; PMID: 28555414). Some pedigrees also show solid-tumor predisposition (PMID: 40858768).
- **Survival/mortality:** No disorder-specific survival statistics are established given rarity; outcomes depend on the specific malignancy that develops and its treatment.
- **Prognostic factors:** Development of malignancy; acquisition of a somatic second hit (loss of wild-type *ETV6*); need for HSCT. No validated prognostic biomarker panel beyond genotype and clinical/molecular surveillance.

12. Treatment

There is no disease-specific or disease-modifying drug for the germline ETV6 defect; management is supportive and surveillance-based (PMID: 28555414).

- **Supportive care for bleeding:** Platelet transfusion and antifibrinolytics as needed for significant bleeding or hemostatic challenges. **ITP-directed therapies (splenectomy, immunosuppression) are inappropriate** and should be avoided — this is not immune-mediated.
- **Surveillance:** Lifelong hematologic monitoring for malignant transformation, with recommendations for patient screening and follow-up (PMID: 28555414).
- **Treatment of malignancy:** Standard protocols for the specific leukemia/MDS, potentially including **allogeneic HSCT** (NCIT: Hematopoietic Cell Transplantation).
- **Carrier-aware donor selection (critical):** For allogeneic HSCT, **related donors must be screened** to avoid using a donor carrying the same germline variant, which risks **donor-derived leukemia** and graft failure (PMID: 36279069: *"Prompt recognition of affected patients is imperative not only for personalized surveillance strategies but also for proper donor selection for those undergoing stem cell transplantation to avoid familial donors who also may share the same germline mutation"*; PMID: 37913908). Empirical risk: of 5 patients

transplanted from a predisposition-variant-positive relative donor, **1 developed donor-derived leukemia** (PMID: 42418371).

- **Genetic counseling:** Integral to management (risk assessment, cascade testing, family planning).
- **Experimental/targeted therapies:** None approved or in trials specifically for the germline ETV6 defect. The **IL-18/IL-13 sterile-inflammation axis** identified in the mouse knock-in is a mechanistic (not yet clinical) therapeutic candidate (PMID: 35803545).
- **Pharmacogenomics:** Not specifically established for this disorder.

NCIT suggestions: Hematopoietic Cell Transplantation, Platelet Transfusion, Supportive Care, Genetic Counseling.

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13. Prevention

- **Primary prevention:** Not possible for a germline disorder. Reproductive options (genetic counseling, preimplantation genetic diagnosis, prenatal testing) can prevent transmission to offspring.
- **Secondary prevention (most important): Surveillance and cascade genetic testing** to detect malignant transformation early and identify at-risk relatives (PMID: 28555414; PMID: 36279069).
- **Tertiary prevention:** Avoid inappropriate ITP therapies; carrier-aware donor selection for HSCT to prevent donor-derived malignancy (PMID: 42418371); appropriate peri-procedural hemostatic management.
- **Genetic counseling:** Central to prevention (autosomal dominant, 50% transmission risk).
- **Immunization / public health / environmental interventions:** Not applicable (no infectious or environmental etiology).

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human *ETV6* (NCBI Gene 2120) has a mouse ortholog *Etv6* (NCBI Gene 14011; NCBI Taxon 10090, *Mus musculus*). ETV6 is evolutionarily conserved as a hematopoietic regulator.

- **Natural disease in other species:** No well-characterized naturally occurring ETV6-related thrombocytopenia syndrome in companion animals or wildlife is documented (no established OMIA entry for this specific germline syndrome). **Not available.**
- **Comparative biology:** The essential role of ETV6 in bone-marrow hematopoiesis is conserved — mouse knockouts demonstrate conserved requirement (Section 15), providing a strong comparative basis even though a spontaneous human-equivalent disease is not reported in other species.
- **Transmission / zoonosis:** Not applicable (germline genetic disorder).

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15. Model Organisms

- **Mouse knockout (constitutive):** *Etv6/Tel*-null mice are **embryonic lethal** due to a yolk-sac angiogenic defect; chimera studies show ETV6 is **essential for bone-marrow hematopoiesis of all lineages**, manifesting within the first postnatal week (PMID: 9694803): *"TEL function is essential for the establishment of hematopoiesis of all lineages in the bone marrow. This defect is manifest within the first week of postnatal life."* and *"TEL-/- mice are embryonic lethal because of a yolk sac angiogenic defect."*
- **CRISPR/Cas9 knock-in mouse (disease allele):** A knock-in of the most common recurrent germline *ETV6* mutation is viable and recapitulates key features — **hematopoietic defects centered on the MPP4 progenitor subset and sterile inflammation** (PMID: 35803545): *"we developed a mouse model of the most common recurrent, disease-causing germline mutation in ETV6. We found defects in hematopoiesis related primarily to abnormalities of the multipotent progenitor population 4 (MPP4) subset ... and evidence of sterile inflammation."* It links **IL-18/IL-13** to impaired B-cell development.
- **Patient-derived iPSC models:** CRISPR-Cas9-engineered and patient-derived iPSC lines recapitulate the megakaryocyte phenotype and distinguish ETV6 from RUNX1 disease (PMID: 34019812): *"The ETV6-mutant iPSCs yield more hematopoietic progenitor cells and megakaryocytes, but the megakaryocytes are immature and less responsive to agonist stimulation"*, and *"both mutant iPSC lines display defects in proplatelet formation ... the molecular mechanisms may be distinct."* A new patient iPSC line, MULi001-A, carries germline p.Pro214Leu (PMID: 42296844).
- **In vitro cell systems:** HeLa and Ba/F3 cells expressing wild-type vs. mutant ETV6 are used for localization and transcriptional-reporter assays (e.g., MMP3, PF4 promoters) to classify variant function (PMID: 42479114; PMID: 35803545).

- **Phenotype recapitulation & limitations:** The knock-in mouse captures progenitor defects and inflammation but is viable (unlike the embryonic-lethal null) and does not fully model human B-ALL incidence; iPSC models capture megakaryocyte maturation/proplatelet defects but lack the in vivo marrow niche and long-term leukemogenesis. Resources: MGI (mouse *Etv6*), Cellosaurus (iPSC lines).

Evidence Base

PMID	Title (abbrev.)	Role in this report
25807284	*Germline mutations in ETV6 … thrombocytopenia, red cell macrocytosis, predisposition to ALL*	Founding gene-discovery paper; triad; p.Pro214Leu/p.Arg418Gly; cytoplasmic mislocalization; dominant-negative mechanism
26102509	*Germline ETV6 mutations confer susceptibility to ALL and thrombocytopenia*	Independent confirmation; impaired nuclear localization; somatic loss of WT allele
27663637	*Germline ETV6 variants … reduced platelet formation, dysfunction, ↑CD34+*	Loss of repressor activity; ETS variants ↓corepressor binding; expanded MK-CFU, defective proplatelet formation
28555414	*Germline ETV6 mutations and predisposition to hematological malignancies*	Clinical overview; THC5; 18 families; malignancy spectrum; surveillance
34019812	*iPSC model of ETV6 vs RUNX1 thrombocytopenia*	ETV6 MKs overproduced-but-immature; shared proplatelet defect; distinct mechanisms
35803545	*Germline ETV6 mutation promotes inflammation …*	CRISPR knock-in mouse; MPP4 defect; IL-18/IL-13; leukemogenesis model
9694803	*TEL/ETV6 required specifically for bone-marrow hematopoiesis*	Normal ETV6 role; KO embryonic lethality
40670159	*Inherited thrombocytopenia with malignancy risk*	10–45% HM risk; impaired aggregation in 90%; ~20-yr diagnostic delay
40858768	*Rare germline ETV6 variant … thrombocytopenia and acute leukemia*	p.Leu376Gln; fibroblast germline confirmation; variable expressivity; solid tumors
36279069	*Germline predisposition syndromes in hematologic malignancies*	Surveillance + carrier-aware donor selection
37913908	*Donor-derived malignancy risks in allo-HSCT*	Screening related donors
42418371	*Germline testing for children undergoing HCT*	1/5 recipients of variant-positive relative donor developed donor-derived leukemia

PMID	Title (abbrev.)	Role in this report
31249973	*Targeted NGS for platelet disorders*	59-gene panel diagnostic modality
42479114	*Functional analysis of rare ETV6 germline variants*	Variant-level functional assays; p.Arg202Gly WT-like vs p.Pro214Leu LOF
28594466	*Bleeding is not the main issue in inherited thrombocytopenias*	Malignancy, not bleeding, dominates concern
42296844	*MULi001-A iPSC line, ETV6 p.Pro214Leu*	Patient iPSC resource

Evidence-type summary: Human clinical/family studies (25807284, 26102509, 27663637, 40670159, 40858768); model organism (9694803, 35803545); in vitro/iPSC (34019812, 42479114, 42296844); clinical-practice/transplant (28555414, 36279069, 37913908, 42418371).

Limitations and Knowledge Gaps

1. **Epidemiology is unquantified.** Only ~18 families were reported by 2017; no reliable prevalence/incidence estimates, no established founder populations, no sex/ethnic distribution data. Carrier frequency unknown (variants absent from gnomAD).
2. **Penetrance estimates are borrowed.** The quoted 10–45% malignancy risk is for the combined ETV6/RUNX1/ANKRD26 group; ETV6-specific, age-stratified penetrance is not precisely defined given small cohorts.
3. **Variant interpretation is difficult.** Rare *ETV6* variants require functional assays; some (p.Arg202Gly) are WT-like despite leukemia co-occurrence, risking over-call of pathogenicity.
4. **The inflammation branch is model-derived.** The IL-18/IL-13 mechanism is demonstrated in the mouse knock-in; its contribution to human thrombocytopenia and leukemogenesis is inferred, not clinically proven.
5. **No therapeutics target the germline defect.** Management is entirely supportive/surveillance-based; no validated prognostic biomarker predicts which carriers will transform.
6. **Modifier genes and epigenetics are unexplored** for this disorder.

Proposed Follow-up Experiments / Actions

1. **Build an international ETV6-RT registry** to generate gene-specific, age-stratified penetrance for B-ALL vs MDS/AML and genotype–phenotype correlations (central-domain vs ETS-domain variants).
2. **Systematic functional classification** of all reported *ETV6* variants using standardized localization + reporter assays (MMP3/PF4 promoters), building an evidence-based pathogenicity framework to reduce VUS and over-calling (PMID: 42479114).
3. **Test the IL-18/IL-13 axis as a therapeutic target** in the knock-in mouse and patient iPSC systems — e.g., IL-18/IL-13 blockade to assess rescue of lymphoid development and thrombopoiesis (PMID: 35803545).
4. **Single-cell and longitudinal clonal tracking** of carrier bone marrow to detect pre-leukemic second hits (wild-type *ETV6* loss) early, enabling risk-adapted surveillance.
5. **Formalize carrier-aware HSCT donor screening** into guidelines and prospectively quantify donor-derived leukemia risk, building on the 1/5 signal (PMID: 42418371).
6. **Define optimal surveillance intervals** (CBC, marrow evaluation) through natural-history study, since no evidence-based schedule currently exists.

Report compiled from 14 confirmed findings and 30 reviewed papers over 5 investigation iterations. All mechanistic and clinical claims are attributed to primary literature with verified PMID citations.
